Boundaries and Defects

in Conformal Field Theories at Finite Temperature

Daniele Artico in collaboration with A. Chalabi

Università di Parma & INFN, Gruppo Collegato di Parma

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D. Artico, A. Chalabi,

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ADAM CHALABI

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Outline

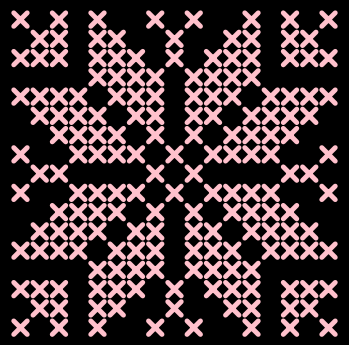
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2 **Thermal defect kinematics**
One-point functions and the new thermal defect data

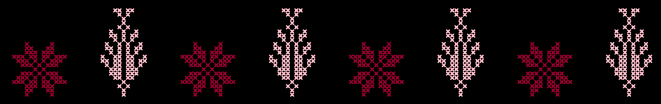
3 **A one-point bootstrap**
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INTRODUCTION



Critical systems with a temperature, a boundary or an impurity are CFTs on non-trivial geometries

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How do the data of a conformal defect enter thermal observables, and which constraints do conformal symmetry and periodicity impose on them?

Finite temperature means a CFT on $S^1_\beta \times \mathbb{R}^{d-1}$: the local data are unchanged, but the geometry adds new data

[El-Showk, Papadodimas, '11][Iliesiu, Koloğlu, Mahajan, Perlmutter, Simmons-Duffin, '18][Marchetto, Miscioscia, Pomoni, '23]

Spectrum and OPE coefficients are those of the CFT on $\mathbb{R}^d$. The scale β allows thermal one-point functions of even-spin primaries:

$$\langle \mathcal{O}^{\mu_1 \cdots \mu_J}_{\Delta}(x) \rangle_\beta = \frac{b_{\mathcal{O}}}{\beta^\Delta} \left(e^{\mu_1} \cdots e^{\mu_J} - \text{traces}\right).$$

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Periodicity of the thermal circle is the KMS condition:

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KMS + OPE is a crossing equation for the thermal data $b_{\mathcal{O}}$.

A conformal defect breaks the conformal group further, and brings its own data

[Cardy, '84][McAvity, Osborn, '95][Billò, Gonçalves, Lauria, Meineri, '16]

A p-dimensional conformal defect of co-dimension $q = d - p$ preserves

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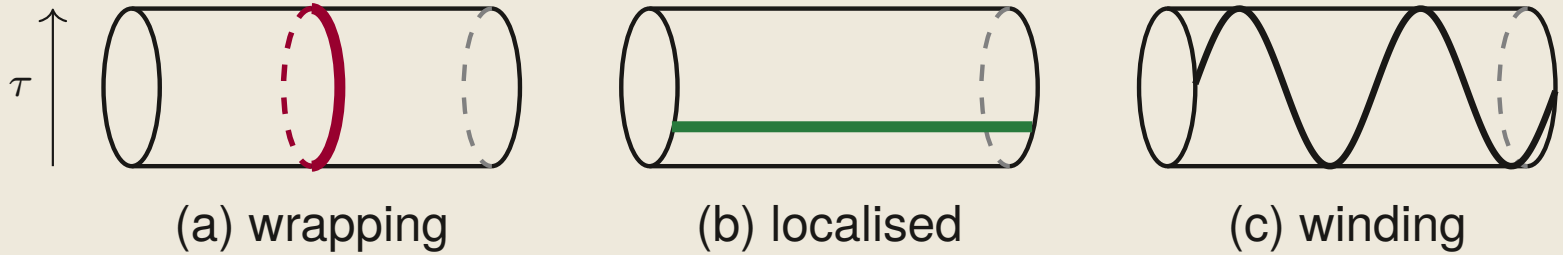
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The DCFT data $\{\hat{\Delta},\ a_{\mathcal{O}},\ \mu_{\mathcal{O}}^{\hat{\mathcal{O}}},\ \hat{\lambda}\}$ fix all defect correlators.

Combining the two, the way the defect sits inside the thermal geometry matters

[Barrat, Fiol, Marchetto, Miscioscia, Pomoni, '24][Li, Nakayama, Nishioka, '26]



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Different embeddings preserve different symmetries.

Same programme as in flat space: symmetry, then data, then bootstrap, then solvable examples

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3) In free theories we extract the thermal defect data and compute thermodynamics: in all examples $|F_{UV}| > |F_{IR}|$.



THERMAL DEFECT KINEMATICS

Wrapping and localised defects preserve different subgroups of the thermal symmetry

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CFT on $\mathbb{R}^d$	$SO(d+1,1)$
defect on $\mathbb{R}^d$	$SO(p+1,1) \times SO(q)$
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For a localised defect, one-point functions can depend on τ.

For a wrapping defect, one-point functions depend on a single cross-ratio that interpolates between defect physics and thermal physics

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Thermal translations are preserved, so a bulk scalar depends only on its distance from the defect, in units of β:

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$F_\mathcal{O}(z)$ interpolates between DCFT data and thermal CFT data.

Defect primaries acquire thermal one-point functions: these are the new thermal DCFT data

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Only transverse scalars with even parallel spin can have

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The $\hat{b}_{\hat{\mathcal{O}}}$ can be read off from $F_{\mathcal{O}}(z)$.

For a boundary, conservation leaves a single free function in the stress tensor

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We classify all tensor structures up to spin 2, for any p and q. Near a boundary ($q = 1$, $p \neq 2$), conservation of $T^{\mu\nu}$ gives

$$\langle T^{\mu\nu} \rangle_\beta = \left(e^\mu e^\nu - \tfrac{1}{d} g^{\mu\nu}\right) \frac{b_T}{\beta^d} + \left((d-1)e^\mu e^\nu + n^\mu n^\nu - g^{\mu\nu}\right) \frac{F_T^{\perp\perp}(z)}{\beta^d} \,.$$

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One function controls the thermodynamics near a boundary.

For a localised defect, KMS already constrains a single one-point function

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Thermal translations are broken, so there are two cross-ratios:

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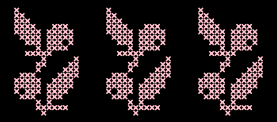
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The simplest correlator with a non-trivial KMS condition is a one-point function.



A ONE-POINT BOOTSTRAP





KMS plus the DOE turns a one-point function into a crossing problem, if the DOE converges far enough

[Hardy, Riesz, '15][**DA**, Chalabi, '26]

Thermal blocks around the defect:

$$F_{\mathcal{O}}(w,z) = \sum_{\hat{\mathcal{O}}} \mu_{\mathcal{O}}^{\hat{\mathcal{O}}} \hat{b}_{\hat{\mathcal{O}}}\, \varrho^{\hat{\Delta}-\Delta} P_s^{(q)}(\eta)\,,$$

with $\varrho = \sqrt{w^2+z^2}$ and $\eta = w/\varrho$.

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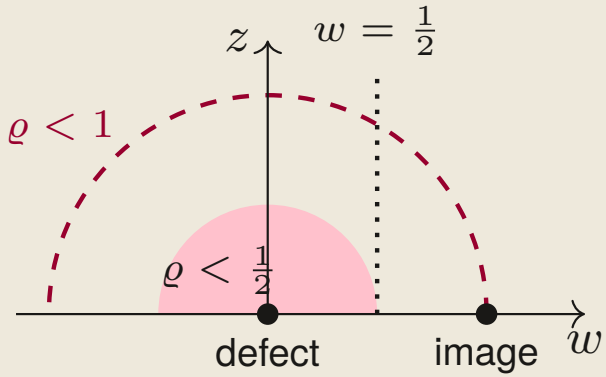
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If $\mu_{\mathcal{O}}^{\hat{\mathcal{O}}} \hat{b}_{\hat{\mathcal{O}}} > 0$ at large $\hat{\Delta}$, Landau's theorem extends convergence to the nearest thermal image, $\varrho < 1$.



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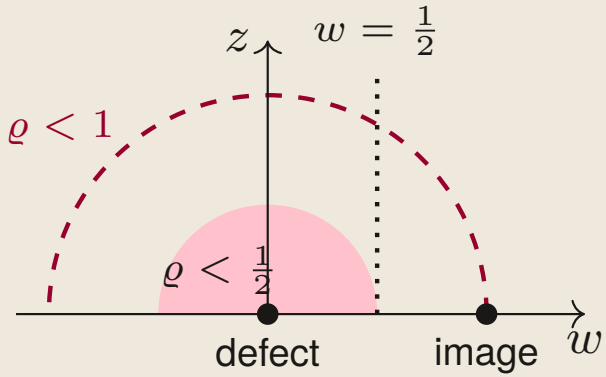
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KMS can then be imposed term by term.

The result is an infinite two-parameter family of sum rules on the thermal defect data

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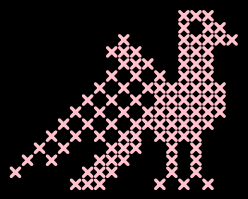
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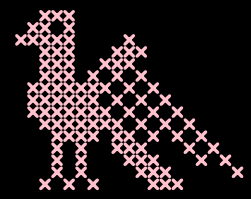
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Periodicity alone gives infinitely many constraints.



FREE-FIELD EXAMPLES





In free theories, thermal images make every observable computable

[Dowker, Kennedy, '78][Kennedy, Critchley, Dowker, '80]

The finite-temperature propagator is a sum over images along the thermal circle:

$$G_\beta(x,y) = \sum_{m\in\mathbb{Z}} (\pm 1)^m \, G(x, y + m\beta e)\,, \qquad + \text{ bosons, } - \text{ fermions.}$$

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Defect	Embedding	Bulk theory
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Monodromy defect	wrapping	complex scalar, $d = 4$
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The examples realise many of the structures allowed by symmetry.

The free scalar BCFT gives infinitely many thermal boundary data in closed form

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Neumann ($\varsigma = +1$) or Dirichlet ($\varsigma = -1$) boundary. From $\langle\, \phi^2 \,\rangle_\beta$:

$$\mu_{\phi^2}^{\hat{\mathcal{O}}} \hat{b}_{\hat{\mathcal{O}}} = \begin{cases} 2(1+\varsigma)\,\zeta(\hat{\Delta})\,\kappa\,, & \hat{\Delta} = d-2\,, \\ -\dfrac{2^{2n+3}\varsigma \cos\left(\pi \frac{d-\hat{\Delta}}{2}\right) \Gamma\left(\frac{\hat{\Delta}}{2}\right) \zeta(\hat{\Delta})}{\Gamma\left(\frac{d-2}{2}\right) \Gamma\left(\frac{\hat{\Delta}-d}{2}+2\right)}\,\kappa\,, & \hat{\Delta} = d+2n\,. \end{cases}$$

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Each $\hat{\Delta}$ belongs to a unique boundary primary, so the $\hat{b}_{\hat{\mathcal{O}}}$ follow by dividing by the known $\mu_{\phi^2}^{\hat{\mathcal{O}}}$.

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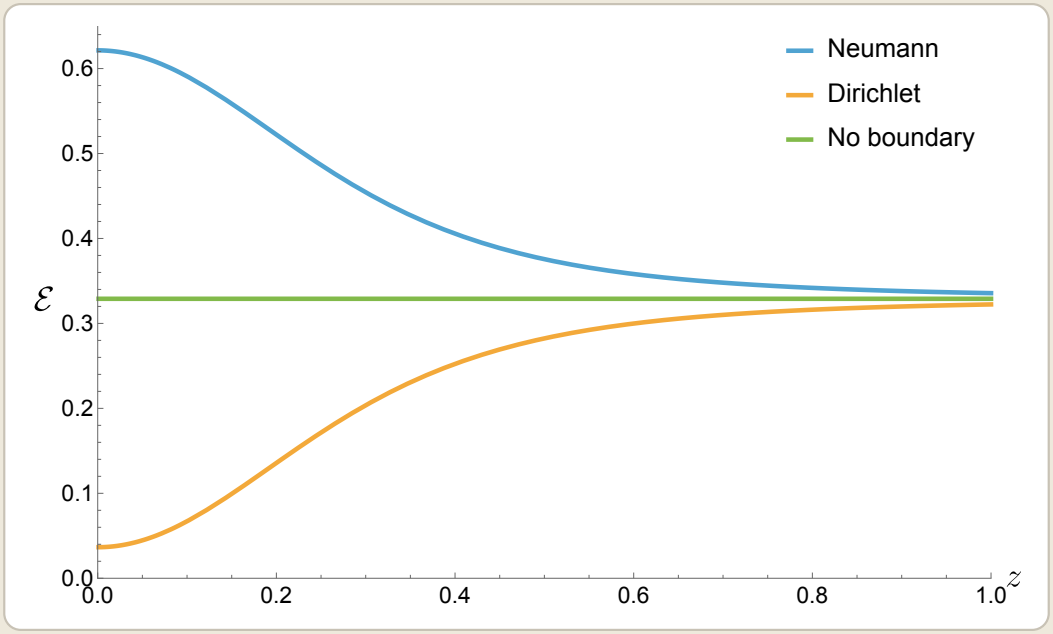
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Every boundary primary in the BOE of ϕ^2 acquires a thermal one-point function.

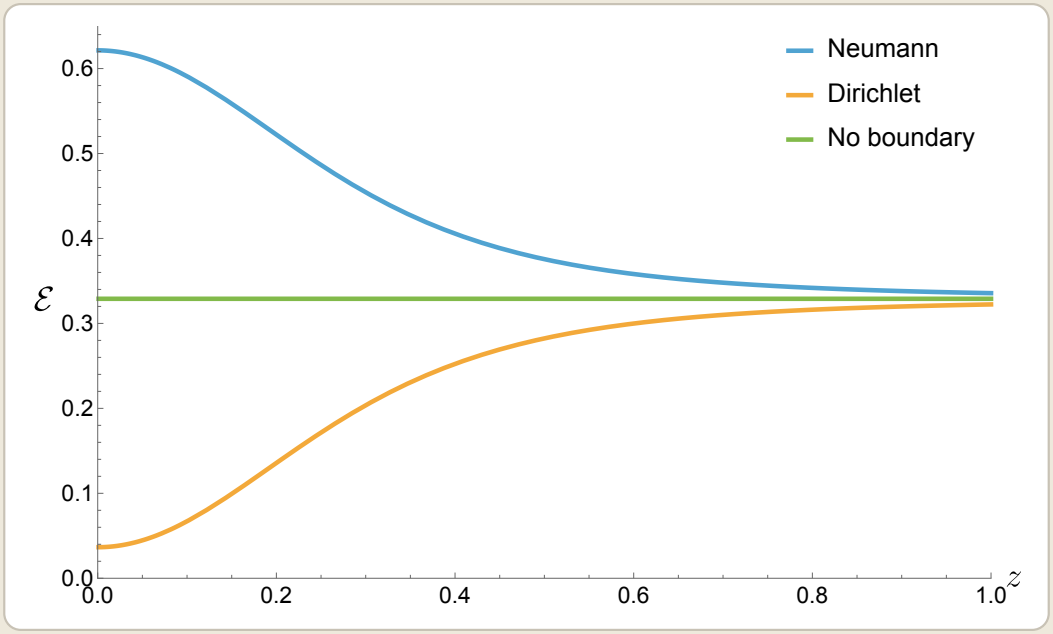
A boundary changes the local energy density.

[**DA**, Chalabi, '26]



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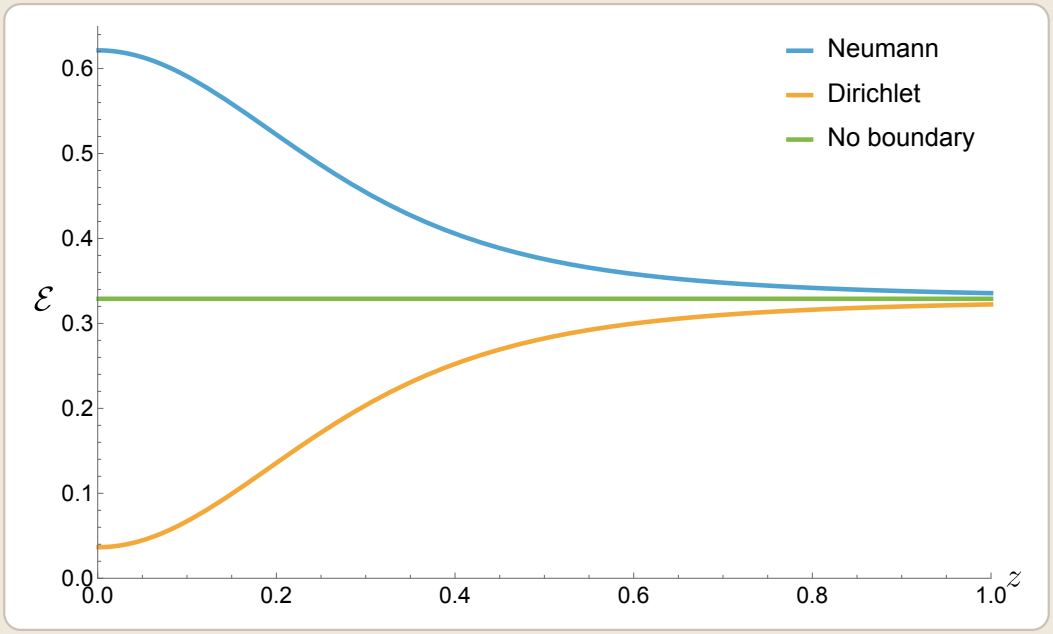
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Free scalar in $d = 4$, $\mathcal{E}(z) = \beta^4 \langle T_{\tau\tau} \rangle_\beta$. At the boundary $\mathcal{E}(0) = \frac{\pi^2}{270}(9 + 8\varsigma) \neq 0$, a thermal effect of $\hat{\Delta} = 4$ boundary operators.

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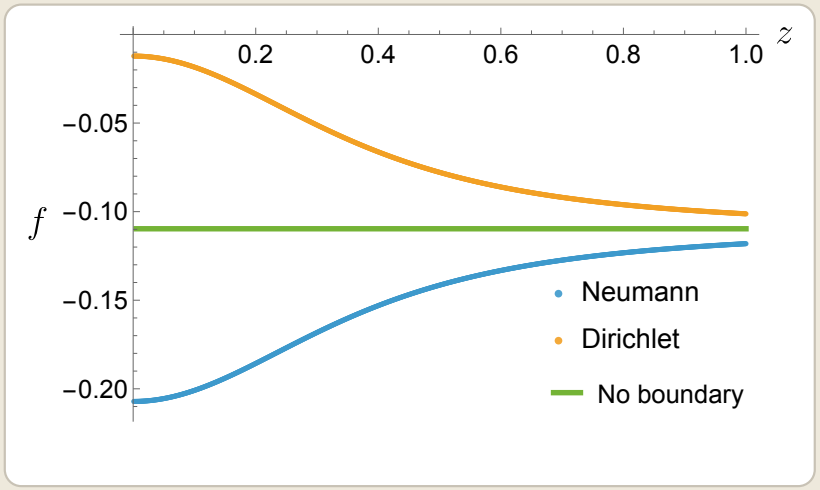


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Neumann has more energy than Dirichlet at every distance

Along the boundary RG flow from Neumann to Dirichlet, the free energy is negative and $|F_{UV}| > |F_{IR}|$

[Appelquist, Cohen, Schmaltz, '99][**DA**, Chalabi, '26]

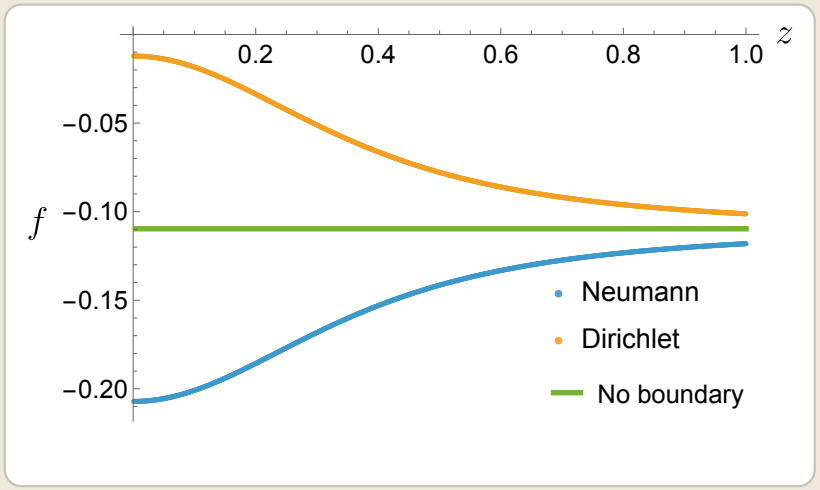


Integrated boundary free energy, for any $d > 2$:

$$F_\partial = -\varsigma \, \frac{A\,\Gamma\left(\frac{d-1}{2}\right)\zeta(d-1)}{4\pi^{\frac{d-1}{2}}\,\beta^{d-1}} \, .$$

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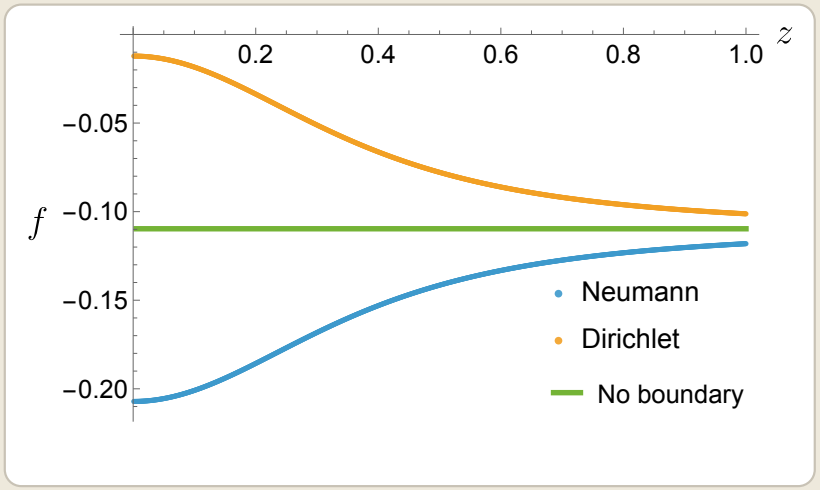
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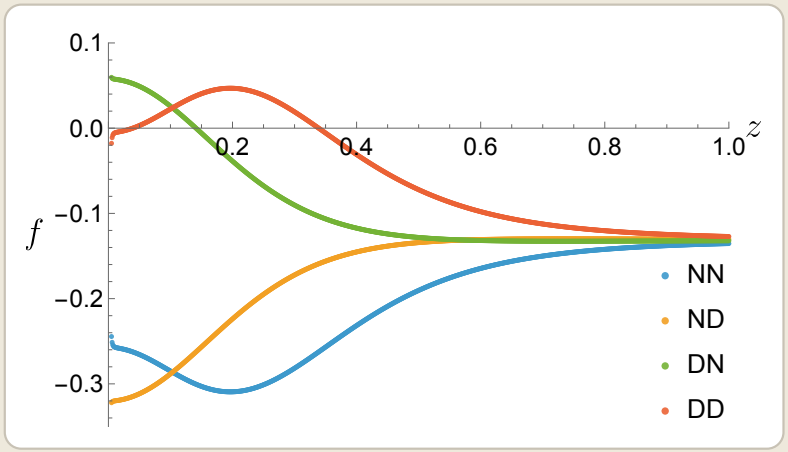
A boundary version of the thermal free-energy bound along RG flows.

In a non-unitary boundary CFT, $|F_{UV}| > |F_{IR}|$ still holds after integrating over space, but not point-wise

[**DA**, Chalabi, '26]

Free four-derivative scalar in $d = 6$, with boundary conditions NN, ND, DN, DD.

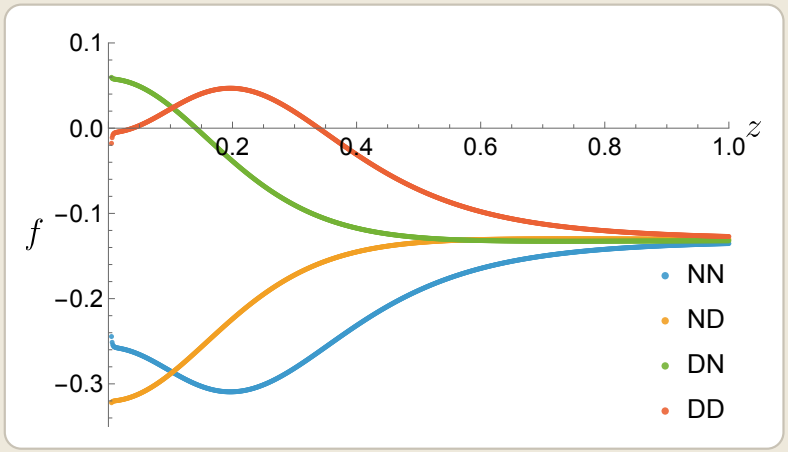
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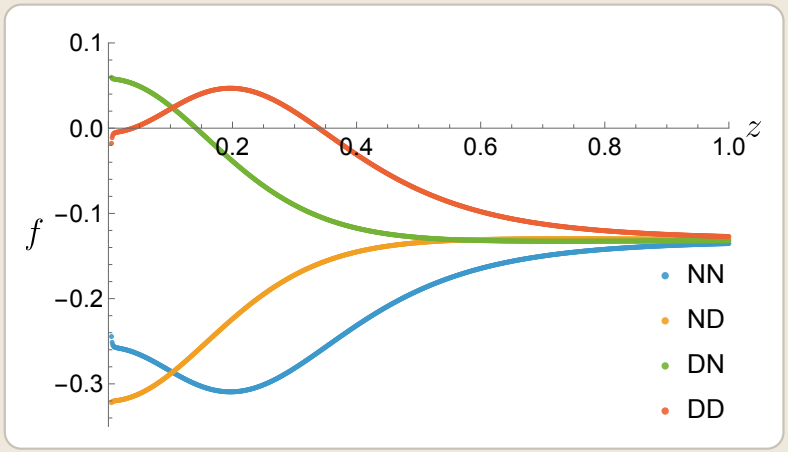
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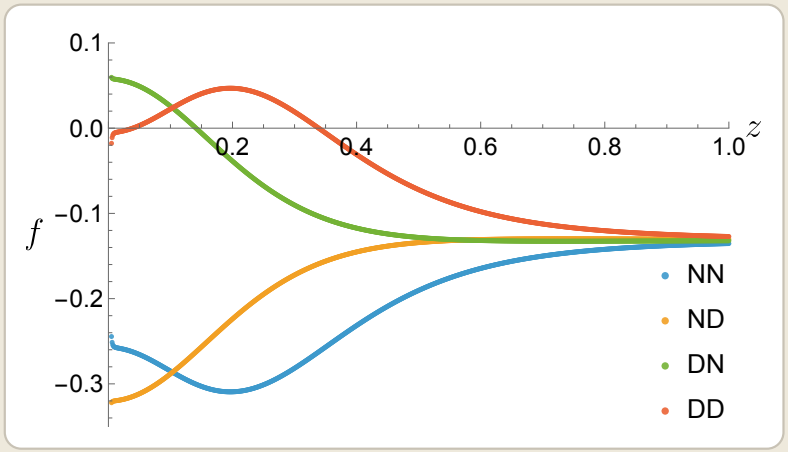
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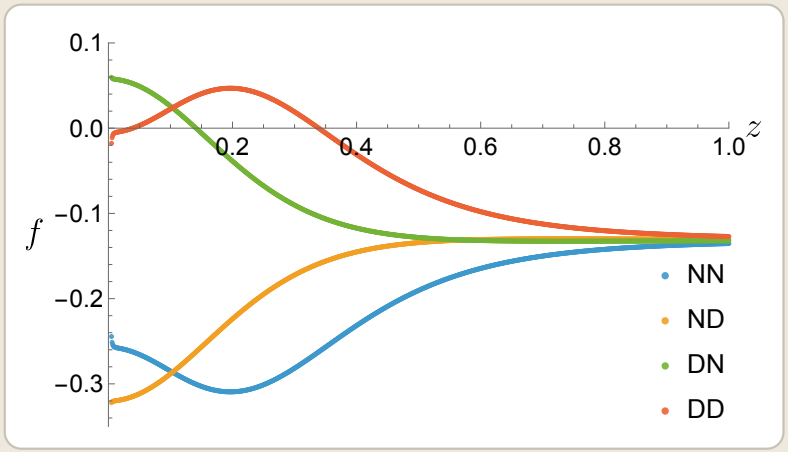
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The integrated inequality survives non-unitarity here.

A monodromy defect has branch points in its one-point functions, even with a free bulk

[Giombi, Helfenberger, Ji, Khanchandani, '21][Bianchi, Chalabi, Procházka, Robinson, Sisti, '21][**DA**, Chalabi, '26]

A co-dimension two defect for the $U(1)$ symmetry of a free complex scalar, wrapping the thermal circle:

$$\Phi(\theta + 2\pi) = e^{-2\pi i\alpha}\, \Phi(\theta)\,.$$

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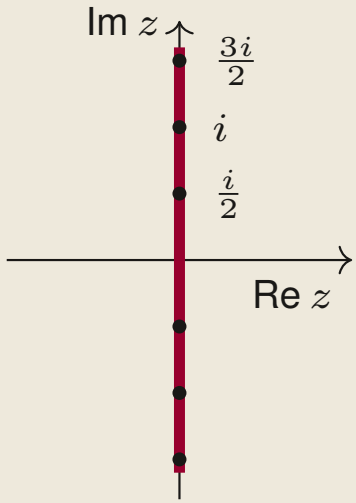
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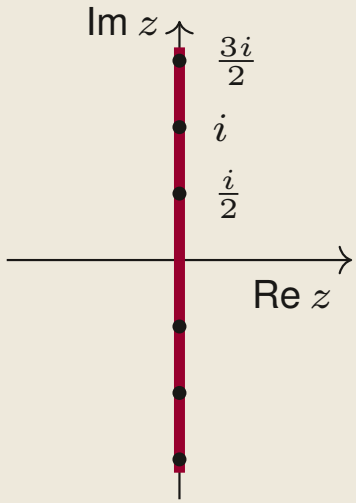
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Non-trivial analytic structure without interactions.

The monodromy defect lowers the thermal energy, and its defect RG flow again satisfies $|F_{UV}| > |F_{IR}|$

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Defect energy in $d = 4$, with ξ the weight of the singular modes:

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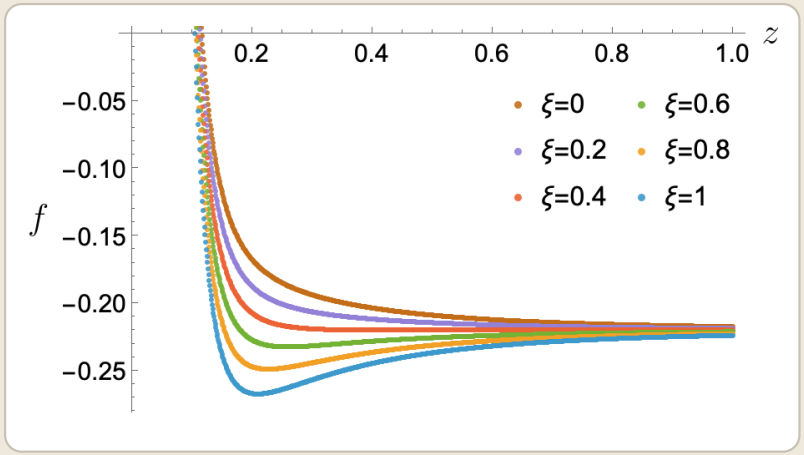
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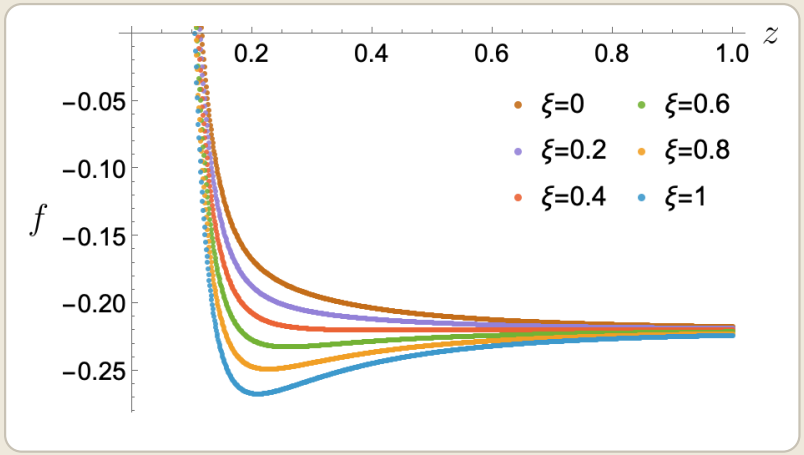
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The same inequality holds for a defect RG flow in co-dimension two.

The localised magnetic line passes every KMS sum rule

[Cuomo, Komargodski, Mezei, '21][**DA**, Chalabi, '26]

$\mathcal{L} = \lambda \int dx_{\parallel}\, \phi(0, x_{\parallel}, \vec{0}_{\perp})$ in the free scalar, $d = 4$, at a point of the thermal circle:

$$\langle\, \phi(x)\, \mathcal{L}\, \rangle_{\beta} = \lambda \kappa \pi \left(\frac{1}{r} + \frac{2}{\beta} \sum_{k \geq 0} \zeta(2k+1)\, \varrho^{2k} P_{2k}(\eta) \right), \qquad \varrho < 1\,.$$

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The DOE converges up to the nearest image, with thermal DCFT data

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All sum rules hold thanks to the identity (R odd)

$$\sum_{k} \frac{\zeta(2k+1)}{4^k} \frac{\Gamma(2k+1)}{\Gamma(2k-R+1)} = \Gamma(R+1)\,.$$

The localised magnetic line passes every KMS sum rule

[Cuomo, Komargodski, Mezei, '21][**DA**, Chalabi, '26]

$\mathcal{L} = \lambda \int dx_{\parallel}\, \phi(0, x_{\parallel}, \vec{0}_{\perp})$ in the free scalar, $d = 4$, at a point of the thermal circle:

$$\langle\, \phi(x)\, \mathcal{L}\, \rangle_{\beta} = \lambda\kappa\pi \left(\frac{1}{r} + \frac{2}{\beta} \sum_{k\geq 0} \zeta(2k+1)\, \varrho^{2k} P_{2k}(\eta) \right), \qquad \varrho < 1\,.$$

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An explicit test of the convergence assumption and of the sum rules.

Its one-point functions decompose into Hurwitz zeta functions, a signature of a dispersion relation

[Barrat, Bozkurt, Marchetto, Miscioscia, Pomoni, '25][**DA**, Chalabi, '26]

Without defects, thermal two-point functions at zero spatial separation decompose as

$$\langle\, \mathcal{O}_1(\tau)\mathcal{O}_2(0)\,\rangle_\beta = \sum_{\mathcal{O}_3} \lambda_{\mathcal{O}_1\mathcal{O}_2}{}^{\mathcal{O}_3} b_{\mathcal{O}_3} \left[\zeta(\Delta_{123}, w) + \zeta(\Delta_{123}, 1-w)\right] + \text{const}\,,$$

with $w = \tau/\beta$ and $\Delta_{123} = \Delta_1 + \Delta_2 - \Delta_3$.

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This suggests a dispersion relation for localised defects.

SUMMARY AND OUTLOOK

Adding a defect to the thermal geometry gives new data, a new bootstrap and new thermodynamics

1) The embedding of the defect in S^1_β decides which constraints exist.

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1) The embedding of the defect in S^1_β decides which constraints exist.

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3) Free examples realise every structure and show $|F_{UV}| > |F_{IR}|$, branch points and Hurwitz zeta decompositions.

Outlook: other geometries

[Benjamin, Lee, Ooguri, Simmons-Duffin, '23][Burić, Russo, Schomerus, Vichi, '24][Burić, Mangialardi, Russo, Schomerus, Vichi, '25, '26]

- Defects on $S^1_\beta \times S^{d-1}$: thermal blocks carry zero-temperature DCFT data, and the high-temperature limit probes heavy defect operators.

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- Defects in the thermal effective action: defect free energies should control the density of defect operators.
- The third embedding: defects winding around the thermal circle.

Outlook: interacting defects and analytic structure

[Cuomo, Komargodski, Mezei, '21][Gimenez-Grau, '22]

× The magnetic line in the $O(N)$ model: test DOE convergence beyond $\beta/2$ and the sum rules in an interacting theory.

Outlook: interacting defects and analytic structure

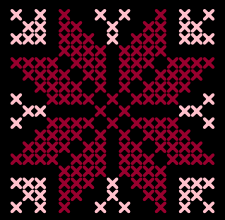
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- × The magnetic line in the $O(N)$ model: test DOE convergence beyond $\beta/2$ and the sum rules in an interacting theory.
- × Derive the dispersion relation behind the Hurwitz zeta decomposition.
- × Real time: quenches, the displacement two-point function and heavy-quark diffusion.





THANK YOU!

